

DreamCampus — Physics Detailed Notes

Concept-focused revision notes for WBJEE, JEE Main & NEET

1. Units, Dimensions & Vectors

- SI base quantities and derived units; dimensional formula; significant figures; absolute, relative and percentage error.
- Scalars have magnitude only; vectors have magnitude and direction. Resolve vectors into perpendicular components before using equations.
- Useful ideas: resultant of perpendicular vectors $R = \sqrt{A^2+B^2}$; dot product $A \cdot B = AB \cos\theta$; cross product magnitude $= AB \sin\theta$.

2. Motion in One Dimension

- Position, distance, displacement, speed, velocity and acceleration. Interpret x-t, v-t and a-t graphs.
- For constant acceleration: $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$.
- Average velocity = total displacement / total time. Average speed = total distance / total time.

3. Motion in Two Dimensions

- Projectile motion is resolved into horizontal and vertical components. Horizontal acceleration is zero when air resistance is neglected.
- Range $R = \frac{u^2 \sin 2\theta}{g}$; maximum height $H = \frac{u^2 \sin^2\theta}{2g}$; time of flight $T = \frac{2u \sin\theta}{g}$ for same launch and landing level.
- Relative velocity is obtained by vector subtraction: $v_{AB} = v_A - v_B$.

4. Laws of Motion

- Newton's first law introduces inertia; second law relates net force and acceleration; third law gives equal and opposite interaction forces acting on different bodies.
- Use free-body diagrams. Common forces: weight mg , normal reaction N , tension T , friction f and spring force kx .
- Static friction adjusts up to its limiting value; kinetic friction is approximately $\mu_k N$ in the basic model.

5. Work, Energy & Power

- Work by a constant force $W = Fd \cos\theta$. Kinetic energy $K = \frac{1}{2}mv^2$. Work-energy theorem: net work = change in kinetic energy.
- Potential energy near Earth's surface $U = mgh$. Mechanical energy is conserved when only conservative forces do work.
- Power is the rate of doing work: $P = dW/dt$; for a constant force and velocity, $P = F \cdot v$.

6. System of Particles & Rotational Motion

- Centre of mass, linear momentum, impulse and conservation of momentum. Impulse equals change in momentum.
- Torque $\tau = rF \sin\theta$. Angular momentum $L = I\omega$ for a rigid body about a fixed axis. Rotational kinetic energy $= \frac{1}{2}I\omega^2$.
- Rolling without slipping: $v = R\omega$. Moment of inertia depends on mass distribution about the chosen axis.

7. Gravitation

- Newton's law: $F = Gm_1m_2/r^2$. Gravitational field is force per unit test mass; potential is work per unit mass with the appropriate reference.
- Orbital speed for a circular orbit around mass M : $v = \sqrt{GM/r}$. Escape speed from radius R : $v_e = \sqrt{2GM/R}$.
- Kepler's laws describe planetary motion; for circular satellites, orbital period increases with orbital radius.

8. Properties of Solids & Fluids

- Stress/strain describe elastic deformation. Young's modulus, bulk modulus and shear modulus quantify elastic response.
- Pressure $P = F/A$; density $\rho = m/V$; buoyant force equals weight of displaced fluid in the ideal Archimedes model.
- Continuity for steady incompressible flow: $A_1v_1 = A_2v_2$. Bernoulli's equation relates pressure, speed and height along a streamline.

9. Thermodynamics & Kinetic Theory

- Temperature, heat, internal energy and work. First law: $\Delta Q = \Delta U + W$ when W is work done by the system.
- Isothermal means temperature constant; adiabatic means no heat exchange in the idealized process.
- Ideal-gas equation $PV = nRT$. Kinetic theory connects microscopic molecular motion with macroscopic pressure and temperature.

10. Oscillations & Waves

- Simple harmonic motion has restoring acceleration proportional to displacement and opposite in direction.
- Spring period $T = 2\pi\sqrt{m/k}$; simple pendulum period for small oscillations $T \approx 2\pi\sqrt{L/g}$.
- Wave relation $v = f\lambda$. Superposition, standing waves, resonance and Doppler effect are important applications.

11. Electrostatics

- Coulomb's law gives inverse-square interaction between point charges. Electric field E is force per unit positive test charge.
- Potential is scalar. For a point charge $V = kq/r$. Equipotential surfaces are perpendicular to electric field lines.
- Capacitance $C = Q/V$; energy stored in a capacitor $U = \frac{1}{2}CV^2 = Q^2/(2C) = \frac{1}{2}QV$.

12. Current Electricity

- Current is rate of flow of charge $I = dQ/dt$. Ohm's law $V = IR$ applies to ohmic conductors under specified conditions.
- Series resistors add; parallel conductance adds. Electrical power $P = VI = I^2R = V^2/R$.
- Kirchhoff's junction and loop rules are used for multi-loop circuits.

13. Moving Charges & Magnetism

- Magnetic force on a charge: $F = qvB \sin\theta$. A current-carrying conductor experiences $F = BIL \sin\theta$.
- Biot-Savart and Ampere laws provide magnetic fields for symmetric current distributions.
- Charged-particle motion in a uniform magnetic field can be circular when velocity is perpendicular to B .

14. Electromagnetic Induction & Alternating Current

- Faraday's law: induced emf is related to the rate of change of magnetic flux; Lenz's law gives the opposing direction.
- Inductors oppose changes in current. In AC circuits, reactance depends on angular frequency.
- Transformers use electromagnetic induction to step voltage up or down in the idealized case.

15. Electromagnetic Waves

- Changing electric and magnetic fields propagate as electromagnetic waves. They can travel through vacuum.
- The electromagnetic spectrum includes radio, microwave, infrared, visible, ultraviolet, X-ray and gamma regions.
- Frequency and wavelength satisfy $c = f\lambda$ in vacuum.

16. Ray & Wave Optics

- Reflection, refraction, total internal reflection, mirrors, lenses and optical instruments form the core of ray optics.
- Thin-lens relation and mirror relation must be used with a consistent sign convention. Magnification depends on object and image distances.
- Wave optics includes interference, diffraction and polarization. Young's double-slit experiment demonstrates interference.

17. Dual Nature, Atoms & Nuclei

- Photon energy $E = hf$. Photoelectric emission depends on photon frequency relative to the material's work function.
- de Broglie wavelength $\lambda = h/p$. Atomic models explain discrete energy levels; nuclear physics covers radioactivity and binding energy.
- Radioactive decay is exponential; half-life is the time for the undecayed population to reduce by half.

18. Electronic Devices

- Intrinsic and extrinsic semiconductors; p-type and n-type materials; p-n junction and diode characteristics.
- Rectification, LEDs, photodiodes, transistors and basic logic/electronic applications are common study areas.
- Focus on device symbol, biasing, characteristic curve and practical use.

How to use these notes

Read one unit, write the core formulas from memory, solve 10–20 objective questions, then revisit mistakes. Always cross-check the current official syllabus before final revision.